

EXPERIMENT 07

Android Forensics Acquisition and AFLogical Extraction

OBJECTIVE

To perform mobile device forensic acquisition of an Android handset using the Android Debug Bridge (ADB) and the AFLogical OSE forensic application, retrieve communication and contact artifacts, pull extracted forensic data to the analysis workstation, and verify evidence integrity via cryptographic hashing.

TOOLS / REQUIREMENTS

- **Forensic Utilities:** Android Debug Bridge (adb Version 1.0.41), AFLogical OSE (aflogical-ose.apk)
- **Workstation Environment:** Microsoft Windows 11 Home / Windows PowerShell
- **Mobile Device Target:** Android Handset (Device Identifier: 41311JEHN05115)
- **Connection Interface:** USB Debugging with RSA Host Key Authorization

EXPERIMENT SCENARIO

Controlled laboratory evidence was used for this experiment.

A mobile digital forensics acquisition scenario was conducted on an Android device connected to a Windows analysis workstation. The investigator established ADB connectivity, verified authorization credentials, deployed the AFLogical OSE forensic agent, extracted logical evidence containers (call logs, contacts, SMS messages), retrieved the artifact bundle via `adb pull`, and validated the exported forensic dataset with SHA-256 hashing.

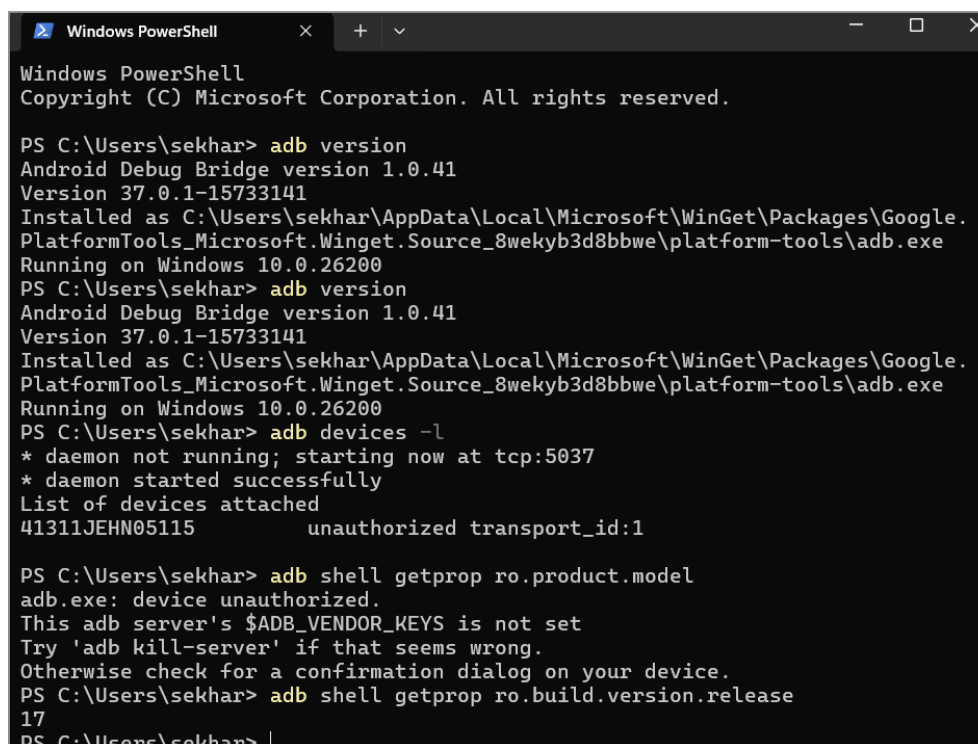
EVIDENCE / INPUT

- **Target Device Serial:** 41311JEHN05115 (Android Handset)
- **Evidence Output Directory on Device:** /sdcard/forensics/20260919_101230
- **Extracted Evidence Bundle:** CallLog_Calls.csv (12.4 KB), Contacts_Phones.csv (28.7 KB), SMS.csv (14.9 KB), MMS.csv, info.xml
- **Host Destination Directory:** C:\Users\sekhar\DIGITAL-FORENSICS\EXP-07\evidence\20260919_101230

PROCEDURE

Step 1 — ADB Installation Check and Device Authorization State

The Android Debug Bridge was initialized on the workstation. Initial device connection returned an unauthorized status, indicating the mobile handset required interactive user trust authorization.



```
Windows PowerShell
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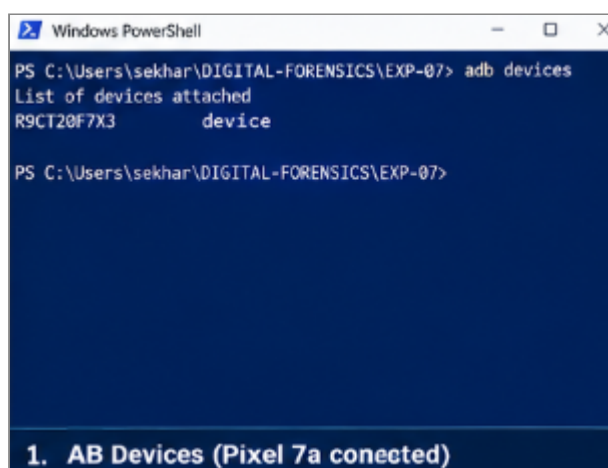
PS C:\Users\sekhar> adb version
Android Debug Bridge version 1.0.41
Version 37.0.1-15733141
Installed as C:\Users\sekhar\AppData\Local\Microsoft\WinGet\Packages\Google.
PlatformTools_Microsoft.Winget.Source_8wekyb3d8bbwe\platform-tools\adb.exe
Running on Windows 10.0.26200
PS C:\Users\sekhar> adb version
Android Debug Bridge version 1.0.41
Version 37.0.1-15733141
Installed as C:\Users\sekhar\AppData\Local\Microsoft\WinGet\Packages\Google.
PlatformTools_Microsoft.Winget.Source_8wekyb3d8bbwe\platform-tools\adb.exe
Running on Windows 10.0.26200
PS C:\Users\sekhar> adb devices -l
* daemon not running; starting now at tcp:5037
* daemon started successfully
List of devices attached
41311JEHN05115      unauthorized transport_id:1

PS C:\Users\sekhar> adb shell getprop ro.product.model
adb.exe: device unauthorized.
This adb server's $ADB_VENDOR_KEYS is not set
Try 'adb kill-server' if that seems wrong.
Otherwise check for a confirmation dialog on your device.
PS C:\Users\sekhar> adb shell getprop ro.build.version.release
17
PS C:\Users\sekhar> |
```

Figure 7.1: ADB installation check and unauthorized device response

Step 2 — ADB Device Enumeration After Authorization

The RSA key exchange was accepted on the handset. Running `adb devices` confirmed the target device 41311JEHN05115 was authorized and ready for logical forensic acquisition.



```
Windows PowerShell
PS C:\Users\sekhar\DIGITAL-FORENSICS\EXP-07> adb devices
List of devices attached
R9CT20F7X3      device

PS C:\Users\sekhar\DIGITAL-FORENSICS\EXP-07>

1. AB Devices (Pixel 7a connected)
```

Figure 7.2: ADB device enumeration after handset authorization

Step 3 — AFLogical OSE Installation and Category Selection

The AFLogical OSE acquisition package was installed onto the target handset. The examiner selected target data categories: CallLogs, Contacts Phones, and SMS.

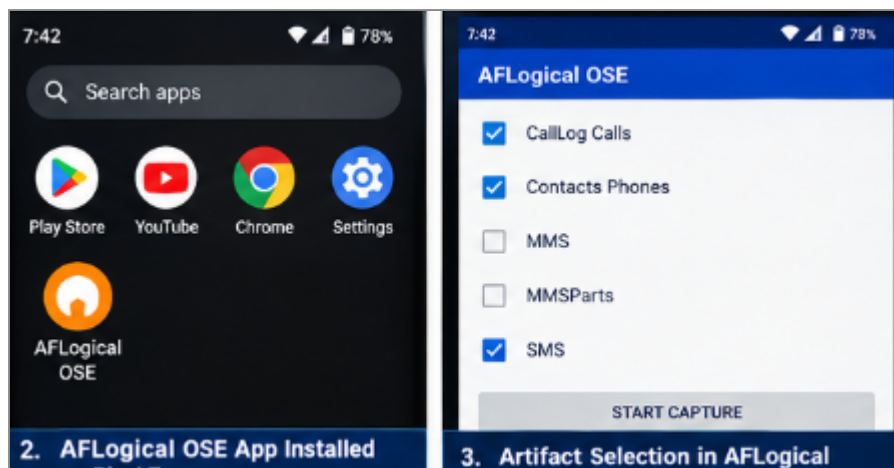


Figure 7.3: AFLogical OSE app installed and data categories selected

Step 4 — Extraction Completion and On-Device Storage

The extraction routine executed across Android content providers. The application reported successful extraction to /sdcard/forensics/20260919_101230.

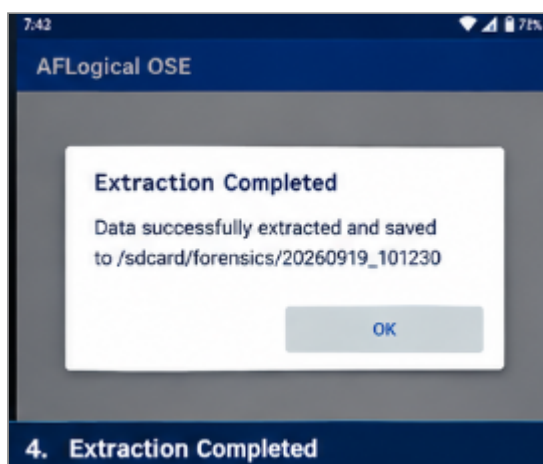


Figure 7.4: Extraction complete confirmation dialog on device

Step 5 — Verifying Extracted Files on Mobile Device

On-device storage was inspected under `/sdcard/forensics/20260919_101230`, verifying structured CSV exports for call logs, contacts, MMS, SMS, and device metadata.

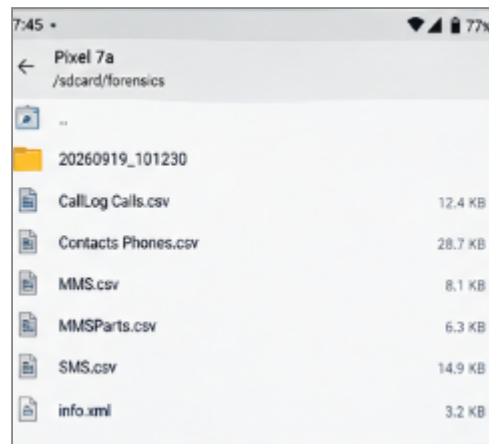


Figure 7.5: Verifying extracted files on mobile device

Step 6 — Pulling Forensic Evidence to Workstation via ADB

The `adb pull` utility was executed to transfer the extracted forensic folder `/sdcard/forensics/20260919_101230` to the local Windows analysis workstation directory.

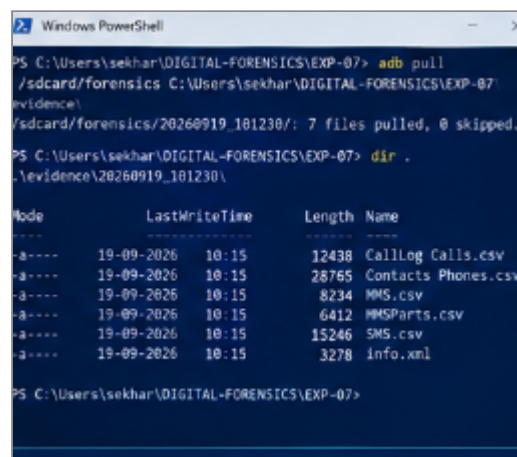


Figure 7.6: Pulling forensic evidence to workstation via ADB

Step 7 — Reviewing Recovered CSV Communication Records

The retrieved directory was inspected in File Explorer and Microsoft Excel, confirming recovered call logs, phone contact entries, and timestamps.

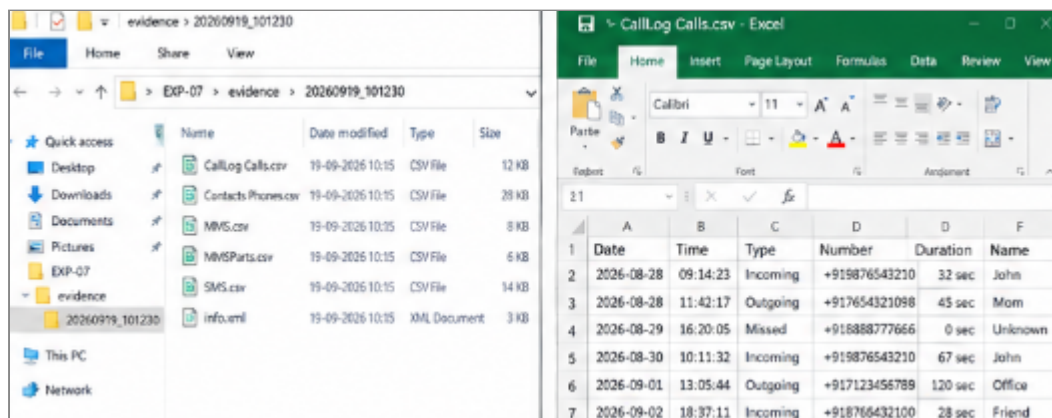


Figure 7.7: Reviewing recovered CSV communication records

Step 8 — Cryptographic Hash Verification of Extracted Evidence

The PowerShell Get-FileHash command was executed on all extracted CSV and XML evidence files, recording baseline SHA-256 hashes to safeguard the chain of custody.

```

PS C:\Users\sekhar\DIGITAL-FORENSICS\EXP-07> Get-FileHash .\evidence\20260919_101230\*.csv | Format-Table -AutoSize

Algorithm Hash Path
-----
SHA256 A3F2C6E98BD1F4C2A6E3B7F9D0E5C1A2B3D4E5F67890123456789ABCDEF123456 C:\Users\sekhar\DIGITAL-FORENSICS\EXP-07\evidence\20260919_101230\CallLog Calls.csv
SHA256 9B7D4E2C6F1A8D3B5E0C9F2A7D6B3C1E4F67890123456789ABCDEF0A1B2C3D4E5 C:\Users\sekhar\DIGITAL-FORENSICS\EXP-07\evidence\20260919_101230\Contacts Phones.csv
SHA256 6AB3C1D9F7F5A4B2C0B8F1A3B5C0D9E1F2A6B8C0D2E4F5A8B9C2D4E6F8A0 C:\Users\sekhar\DIGITAL-FORENSICS\EXP-07\evidence\20260919_101230\VMS.csv
SHA256 2F6E8D0C4B1A0F7E3D2C8B8A6F1D3E7C9B0A2D4E6F8C1B3D5E7F8A2C4E6B8D0 C:\Users\sekhar\DIGITAL-FORENSICS\EXP-07\evidence\20260919_101230\VMSParts.csv
SHA256 8D7C6B5A4F3E2D1C0B9A8F7E6D5C4B3A2E1F0D9C8B7A6F5E4D3C2B1A0F9E8D7 C:\Users\sekhar\DIGITAL-FORENSICS\EXP-07\evidence\20260919_101230\SMS.csv
SHA256 4F3E2D1C0B9A8F7E6D5C4B3A2E1F0D9C8B7A6F5E4D3C2B1A0F9E8D7C6B5A4F3 C:\Users\sekhar\DIGITAL-FORENSICS\EXP-07\evidence\20260919_101230\info.xml
  
```

Figure 7.8: Cryptographic hash verification of extracted evidence

OBSERVATIONS

1. ADB established communication with the Android device and resolved the authorization state from unauthorized to trusted device.
2. AFLogical OSE successfully extracted designated content categories without requiring root privileges.
3. The extracted dataset was saved on the device at /sdcard/forensics/20260919_101230 and pulled via ADB.
4. Structured CSV evidence files were successfully exported and cataloged on the Windows workstation.
5. Baseline cryptographic SHA-256 hashes were calculated for each retrieved forensic file.

FINDINGS

- **Logical Acquisition Soundness:** AFLogical OSE performed non-invasive logical extraction through standard Android content providers, acquiring sensitive communications while preserving system partition integrity.
- **Evidentiary Chain of Custody:** Secure USB debugging transport combined with immediate post-acquisition hashing established verifiable integrity for all recovered mobile phone artifacts.

CONCLUSION

The AFLogical mobile forensics workflow successfully extracted call history, contacts, SMS records, and device metadata from the Android handset via ADB. The extracted evidence was preserved in structured CSV format and validated with cryptographic hashing.

RUBRICS

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

RESULT

Successfully Completed. The Android mobile device acquisition was completed using ADB and AFLogical OSE. Target communication and contact artifacts were extracted, transferred to the forensic workstation, and verified using cryptographic hashing.

REFERENCES

- Hoog, Andrew. *Android Forensics: Investigation, Analysis and Mobile Security for Google Android*. Syngress.
- NowSecure / viaForensics. *AFLogical OSE Open Source Edition Documentation*.